

When Isotropy Fails: Selecting a JEPA’s Collapse-Avoidance Geometry from the Data Distribution

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Abstract

Joint-embedding predictive architectures (JEPAs) avoid representational collapse with a *geometry*: an exponential-moving-average (EMA) target, a variance–covariance penalty (VICReg), or a push toward an isotropic-Gaussian embedding (LeJEPA/SIGReg), and it is almost always fixed a priori and applied across domains. Recent theory undercuts that practice: pushing embeddings toward an isotropic Gaussian recovers the world’s latents *iff* those latents are Gaussian (within a class of worlds), no fixed covariance shape is universally optimal, and global isotropy does not reliably track representation quality. Yet the field only argues *against* a fixed geometry; nobody measures the deciding property and *selects* the geometry from it. We introduce a training-free selector: the embedding distribution’s sliced-negentropy non-Gaussianity, validated by a column-shuffle permutation null that certifies the signal reflects genuine joint structure rather than a sampling or random-encoder artifact, and use it to choose among {EMA, frozen, VICReg, isotropic-Gaussian}. In controlled synthetic data the rule sign-flips as predicted: isotropic-Gaussian is best when latents are Gaussian (rank 1/4) and worst when they are clustered (rank 4/4), with the selector statistic tracking the crossover. On a real genome-JEPA, where genomes read non-Gaussian in every representation we test, the geometry bake-off ranks as the selector predicts (EMA 0.712, VICReg 0.703 vs. SIGReg 0.684) under a *faithful* characteristic-function SIGReg with the loss weight tuned per geometry; SIGReg no longer trails the random-feature floor, so the ranking is not an artifact of a non-faithful or over-weighted regularizer. The gap is small and rests on three seeds, so we assert a ranking rather than a large effect, and we do not obtain a naturally-Gaussian real control (CIFAR-10 in a pretrained encoder is itself non-Gaussian). We conclude that selecting collapse-avoidance geometry from a measured statistic is a promising, testable principle, and provide a cheap permutation-validated rule for it.

1 Introduction

Self-supervised world models learn by predicting in representation space rather than in pixel or token space [1–3]. Their central technical hazard is *representational collapse* (the encoder mapping every input to a near-constant), and every method adopts some *collapse-avoidance geometry* to prevent it: an EMA / stop-gradient target [2, 4, 5], a variance–covariance penalty [6], or, most recently, an explicit push toward an isotropic-Gaussian embedding distribution [7]. This geometry is almost always *fixed a priori*, chosen once and applied across datasets and domains as if one shape were universally right.

Recent theory says that is a mistake. Klindt et al. [11] prove that the isotropic-Gaussian objective (LeJEPa/SIGReg [7]) yields a linearly identifiable representation *iff* the world’s latent variables are Gaussian, with the converse ruling out non-Gaussian alternatives. Importantly, this guarantee is stated within a class of worlds (latents evolving under stationary, additive-noise transitions), so it is data-conditional rather than universal: whether the objective recovers a faithful representation depends on the data-generating process, not the objective alone. A companion line argues that no single fixed covariance shape is universally optimal and instead adapts the target shape via a Sliced-Wasserstein sketch [12]. And an empirical study of vision encoders finds that *global* geometric regularity such as isotropy does not track capability, which is instead predicted by a functional-sensitivity (input–output Jacobian) measure [13]. Taken together, these results suggest that fixing the embedding geometry a priori, and fixing it to a “round Gaussian cloud” in particular, is not obviously the right default.

Yet the field has stopped at *arguing against* a fixed geometry. Nobody has closed the loop: *measure the data property the theory says is decisive, and use it to select the geometry*. That is the gap we close. We build a training-free, validated selector for the deciding property (the embedding distribution’s non-Gaussianity), show in controlled synthetic data that it predicts which of {EMA, frozen, VICReg, isotropic-Gaussian} wins, with a clean sign-flip as the data moves Gaussian $\rightarrow$ clustered, and then test the prediction on a real genome-JEPa, where its representations read non-Gaussian. On genomes the geometry bake-off ranks as predicted, and this holds under a faithful characteristic-function SIGReg with the loss weight tuned per geometry, so the ranking is not an artifact of a non-faithful or over-weighted regularizer; the non-Gaussian reading persists at every taxonomic scale we measure.

Contributions.

1. **A validated, training-free geometry selector.** Sliced-negentropy non-Gaussianity in a data-reflecting coordinate, gated by a column-shuffle permutation null that proves the signal is genuine joint structure, correcting an earlier central-limit-theorem (CLT)-dominated selector that read Gaussian for real and random inputs alike (§3.2, §4.2).
2. **A clean predictor validation.** In controlled synthetic latents the selection rule sign-flips: isotropic-Gaussian is best when the data is Gaussian and worst when it is clustered, with the selector statistic tracking the crossover (§4.1).
3. **A real-data test that survives the obvious confounds.** On the genome-JEPa, EMA/VICReg outrank isotropic-Gaussian on held-out-family trait recovery, as the selector predicts, and the ranking holds under a *faithful* characteristic-function SIGReg with per-geometry loss-weight tuning, with SIGReg no longer below the random-feature floor (§4.4).
4. **A domain characterization.** Genome representations read non-Gaussian at every taxonomic scale we measure, so the reading is not a fluke of one breadth (§4.2).

2 Related Work

Deriving inductive bias from the data. Geometric Deep Learning [8] is the closest *named* principle: derive architecture and inductive bias from the data’s symmetry and geometry. We differ on the lever: a measured *distributional statistic* (non-Gaussianity) rather than symmetry, applied to the collapse-avoidance geometry specifically. The union-of-manifolds / varying-intrinsic-dimension line [9] is the closest actual *move*: measure a data property, design a matching inductive bias. We

extend this from a single statistic on one axis to *selecting* a JEPA collapse geometry, tied to an identifiability condition.

Collapse-avoidance geometry. EMA/stop-gradient methods [4,5], VICReg [6], and LeJEPA/SIGReg [7] all fix their geometry a priori. LeJEPA proves the isotropic Gaussian is the unique optimal embedding distribution and enforces it with SIGReg, using random one-dimensional projections and characteristic-function matching; it is the default we probe on genomes. A complementary line [12] argues no fixed shape is optimal and fits an *arbitrary* target covariance via a Sliced-Wasserstein sketch; that is a different move: it fits a continuous shape and needs a downstream signal, whereas we make an *upstream, training-free selection among discrete, interpretable geometries* from a measured statistic, connected to the Gaussian identifiability condition.

Is isotropy even the goal? A study of vision encoders finds that global geometric regularity such as isotropy does not track compositional capability, which is better predicted by a functional-sensitivity (input–output Jacobian) measure [13]. We read this narrowly: global-geometry scores are not a reliable proxy for representation quality. Accordingly we do not treat isotropy as a target or measure geometry as a quality score; our claim is only that *forcing* isotropy can destroy task-relevant structure when the data is not Gaussian, and our outcome is downstream generalization, not a geometry statistic.

3 Method

3.1 The geometry-selection principle

Let a JEPA encoder f_θ produce embeddings $z = f_\theta(x) \in \mathbb{R}^d$ over a data distribution $\mathcal{D}$, and let collapse be prevented by a geometry $g \in \mathcal{G} = \{\text{EMA}, \text{FROZEN}, \text{VICREG}, \text{SIGREG}\}$. Klindt et al. [11] show, within their class of worlds, that the isotropic-Gaussian geometry (SIGREG) attains a linearly identifiable representation iff the world’s *latent* distribution is Gaussian. This motivates a *selector* s that measures how far a data-reflecting representation is from Gaussian, together with a rule: prefer a shape-agnostic geometry (VICREG/EMA) when s is large, and the isotropic-Gaussian geometry when $s \approx 0$.

What s measures, and what it does not. The identifiability condition is a statement about the world’s *latents*, which relate to observations through an unknown nonlinear mixing. Non-Gaussian observations are in general compatible with Gaussian latents (this is the standard setting of nonlinear ICA), so an observation distribution does not by itself pin down the latent one. We therefore do not claim to measure the latents. We compute s on several *representations* of the genome (pretrained-encoder features, §4.2) as a proxy, and treat non-Gaussianity that appears consistently across independent representations as evidence for, not proof of, non-Gaussian structure the geometry must contend with. Establishing that this proxy implies the latent condition is an open problem we do not solve here; our decisive test (§4.4) instead checks the rule’s prediction directly, on held-out generalization. We did also compute s on the JEPA’s *own* embeddings, as this definition literally asks: the trained embeddings read non-Gaussian for every geometry and, tellingly, *least* for SIGREG ($z=13$ vs. 35 for FROZEN), confirming SIGReg does compress toward Gaussian. But this cannot serve as a *pre-training* selector: on trained embeddings it is circular, since the geometry being chosen partly sets the embedding shape, and on untrained frozen-init embeddings it is central-limit-Gaussianized. This is exactly why we read s off external, data-reflecting representations instead.

3.2 A training-free non-Gaussianity selector

Given an embedding batch $z \in \mathbb{R}^{N \times d}$, we symmetrically whiten the covariance, project onto M random unit directions u_m , and average the *sliced negentropy*, a Hyvärinen entropy proxy [10]:

$$s(z) = \frac{1}{M} \sum_{m=1}^M \left[k_1 (\mathbb{E}[v e^{-v^2/2}])^2 + k_2 (\mathbb{E}[e^{-v^2/2}] - \sqrt{\frac{1}{2}})^2 \right], \quad v = z \widehat{u}_m, \quad (1)$$

where v is the whitened, standardized projection and k_1, k_2 are the standard Hyvärinen constants. This is ≈ 0 for any Gaussian (of any covariance) and positive for heavy-tailed *or* multimodal/clustering z . We use it in place of Mardia kurtosis, which is blind to symmetric multimodality (clusters are platykurtic).

Validity control. A raw statistic cannot distinguish genuine joint structure from a per-axis or encoder artifact. For each coordinate we therefore build a null by *column-shuffling*: independently permuting each feature across samples, which preserves every marginal exactly but destroys joint structure. We recompute s on $B=200$ shuffles, and report $z = (s_{\text{real}} - \mu_{\text{null}})/\sigma_{\text{null}}$ and a permutation p . A coordinate is *validly non-Gaussian* iff it beats its own shuffle null ($z > 3$, $p < 0.01$). When $d \gg N$ we PCA-reduce via the $N \times N$ Gram matrix, which is identical to the SVD reduction in the whitened space the statistic operates in but far cheaper. This control is what an earlier frozen-random-init selector lacked: a random convolutional encoder with mean-pooling Gaussianizes any input by the CLT, so that selector read Gaussian for real genomes *and* random amino-acid sequences alike, uninformative.

Coordinates. ESM-2, KEGG-module, and eggNOG features are precomputed genome-level matrices. For the **Evo2** coordinate we sample 8 DNA windows of 4kb per genome, embed each with Evo2 1B (bf16, layer `blocks.24.mlp.13`, hidden 1920; FP8 input projections disabled to run on non-Hopper hardware), and mean-pool over sequence length and windows to a genome vector, then score as above ($N=675$, PCA to 40 dims since $d \gg N$).

3.3 The genome-JEPA and the four geometries

Each protein (amino-acid sequence, 20 residues+pad, truncated at 512) is embedded by a learnable convolutional gene encoder (a 32-d residue embedding, three `Conv1d(k=7, stride 2)` blocks with GELU and GroupNorm (channels 32→64→128→256), global mean-pool, and a linear head) yielding a 256-d gene embedding (0.63M trainable parameters; no tokenizer, no pretrained expert). A genome embedding is the mean of its gene embeddings. The objective is *evolutionary-conditioning JEPA*: a predictor MLP takes a gene’s embedding concatenated with the pair’s divergence (percent-identity/100) and predicts the LayerNorm’d latent of an ortholog in another genome, under a Smooth-L1 loss with a stop-gradient target. All geometries share the encoder, predictor, objective, data, and hyperparameters; they differ only in collapse avoidance:

- EMA: target is an EMA copy of the encoder (decay 0.996); no regularizer.
- FROZEN: target is a frozen random-init encoder; no regularizer (a random-feature floor, not a fair frozen-expert).
- VICREG: live target; $\text{ReLU}(1 - \sqrt{\text{Var } z})$ variance hinge plus off-diagonal covariance penalty $\|\text{offdiag}(\text{Cov } z)\|^2/d$.

- SIGREG (LeJEPA): live target; a faithful characteristic-function normality penalty. For random unit directions w , we push each 1-D projection $u = zw$ toward $\mathcal{N}(0, 1)$ with the differentiable Epps–Pulley / BHEP statistic $T = \frac{\exp(-\beta^2(u_j - u_k)^2/2)}{2(1+\beta^2)^{-1/2} \exp(-\beta^2 u^2/2(1+\beta^2))} + (1+2\beta^2)^{-1/2}$, averaged over directions. By Cramér–Wold this constrains *all* moments toward $\mathcal{N}(0, I)$, not just the first two, matching the published SIGReg rather than a covariance-plus-kurtosis surrogate.

Ortholog pairs from ~ 675 bacterial genomes are sampled to 120,000 triples; we train with AdamW (lr 3×10^{-4} , weight decay 0.05), batch 128, gradient clip 1.0. The powered sweep is all four geometries $\times 3$ seeds $\times 40$ epochs. Parallel data loading (num_workers=24) yields 7.2s/epoch on one A10 GPU (a $22\times$ speedup over a single-worker baseline); the full sweep is ~ 1.5 h at $\sim$ US\$2–3.

3.4 Evaluation

We measure *held-out-family trait recovery*: a logistic probe on the frozen genome embedding, scored by mean AUROC over binarizable BacDive traits on genomes whose *family* is unseen in training: the out-of-taxon generalization used throughout the series. The bake-off reports mean held-out-family AUROC per geometry, averaged over seeds.

4 Experiments

4.1 Synthetic sign-flip: the selection rule, validated

We dial true latents $z = (1-\alpha) \cdot \text{Gaussian} + \alpha \cdot (\text{cluster-center} + \text{heavy tail})$ for $\alpha \in \{0, .25, .5, .75, 1\}$, nonlinearly scramble them into observations, and train four tiny JEPAs differing only in geometry (3 seeds). Metric: linear identifiability R^2 (Ridge readout of true latents). Table 1 and Figure 1 show a clean crossover: the measured non-Gaussianity rises monotonically ($-0.1 \rightarrow 118.4$), and the isotropic-Gaussian geometry moves from best (rank 1/4 at $\alpha=0$) to worst (rank 4/4 at $\alpha=1$), while VICReg (which decorrelates but does not force a shape) moves from middling to best. The selection rule holds across the full range.

Table 1: Synthetic identifiability R^2 (3 seeds). The winner flips with the data shape; the selector statistic tracks it. Best per row in **bold**.

α	non-Gaussianity	frozen	ema	vicreg	sigreg (iso-Gaussian)	winner
0.00	−0.1	0.493	0.417	0.455	0.557	sigreg
0.25	2.2	0.534	0.530	0.538	0.533	vicreg
0.50	41.2	0.644	0.742	0.758	0.519	vicreg
0.75	102.2	0.686	0.752	0.833	0.432	vicreg
1.00	118.4	0.684	0.763	0.836	0.471	vicreg

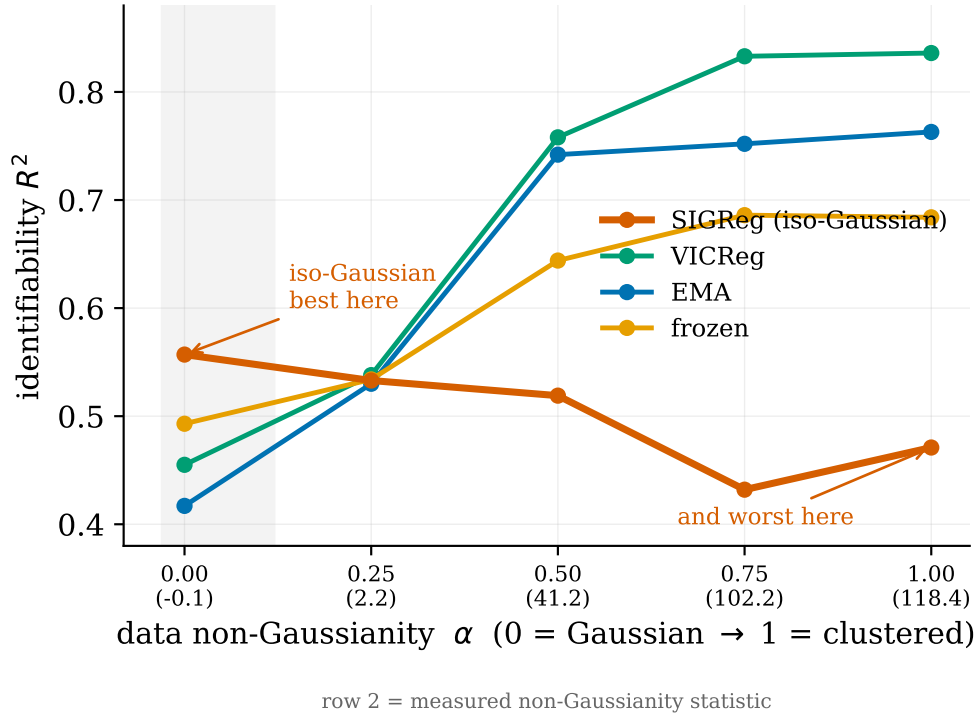


Figure 1: **The selection rule sign-flips.** Identifiability R^2 vs. data non-Gaussianity (synthetic, 3 seeds). The isotropic-Gaussian geometry (SIGReg, thick) is best when the data is Gaussian ($\alpha=0$) and worst once it is clustered, while the shape-agnostic geometries rise. The measured non-Gaussianity statistic (x-axis, in parentheses) tracks the crossover.

Figure 1 makes the mechanism visual. The thick isotropic-Gaussian curve begins highest and ends lowest, crossing the three shape-agnostic curves as the data leaves the Gaussian regime; forcing embeddings toward a round Gaussian helps recovery when the latents really are Gaussian, but actively destroys the cluster structure that carries the signal when they are not. No fixed geometry is best everywhere, and the winner is a deterministic function of a property we can measure from the data before training. The remaining experiments ask where real genomes fall on this axis, and whether the geometry the selector prescribes actually wins.

4.2 Genomes are non-Gaussian at every scale

We run the validated selector on four data-reflecting representations of real genomes (Table 2): a protein language model (ESM-2), machinery space (KEGG modules), functional composition (eggNOG orthologous groups), and the genomic foundation model Evo2 (1B, mean-pooled DNA-window embeddings). Because the sliced-negentropy estimator carries a finite-sample bias that grows with N and d , we score all four at *matched* sample size ($N=675$, Evo2’s size) and matched dimension ($d=40$ after PCA); an earlier unmatched version inflated the apparent spread and is not used. All four are decisively non-Gaussian: each real value exceeds its own marginal-preserving null by $z = 25$ to 104. We deliberately do *not* rank the representations against one another. The null level itself differs across them (representations with heavier-tailed principal components have a higher shuffle baseline, e.g. Evo2’s null of 0.021 vs. eggNOG’s 0.001), so raw statistics are not comparable row to row. The robust, honest claim is only that genome structure reads non-Gaussian

in every representation we examine, including a state-of-the-art genomic FM.

Table 2: Non-Gaussianity of genome representations at matched $N=675$, $d=40$ vs. a column-shuffle null ($B=200$). All four are decisively non-Gaussian ($z \gg 3$). We do not compare raw values across rows: the null level varies with each representation’s marginals, so only the gap to each row’s *own* null is interpretable.

representation	real	null mean	null sd	z
Evo2 1B (genomic FM)	0.0390	0.0210	0.0007	26.0
eggNOG (composition)	0.0129	0.0014	0.0001	103.7
ESM-2 (protein-LM)	0.0048	0.0012	0.0001	39.1
KEGG (machinery)	0.0032	0.0010	0.0001	24.7

How do we know these readings reflect genuine joint structure rather than heavy per-axis marginals or a small sample? The null control is plotted in Figure 2. For each representation we recompute the statistic on 200 column-shuffles that hold every feature’s marginal fixed but scramble the joint structure across features. If the apparent non-Gaussianity came only from per-axis shape or finite-sample noise, the real value would fall inside this shuffle null. It does not: in all four the real statistic (orange) lands far in the right tail of its own null (grey), by 25 to 104 standard deviations. This is precisely the test the earlier random-encoder selector failed, where real genomes and random amino-acid sequences read identically because a random encoder Gaussianizes any input. One residual limitation: the shuffle null has diagonal covariance, so its whitening step is trivial, whereas the real data requires estimating and inverting a full covariance from finite samples; matching that estimation error in the null (for example by shuffling within a decorrelated basis) would tighten the test further, and we leave it to future work.

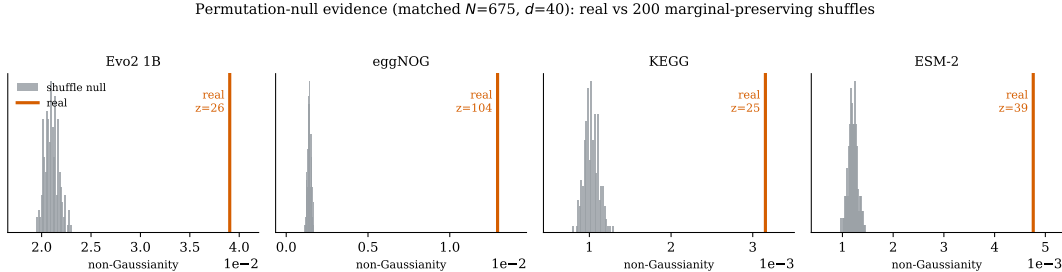


Figure 2: **Permutation-null evidence (the rigorous test), at matched $N=675$, $d=40$.** For each representation, the real sliced-negentropy statistic (orange) vs. 200 column-shuffle nulls (grey) that preserve every marginal but destroy joint structure. The real value sits far in the right tail of its own null in all four ($z = 25$ – 104), so the signal is genuine joint structure, not a per-axis or sampling artifact. A truly Gaussian representation would land *inside* its null; this is what the earlier random-encoder selector failed. Note the null level itself varies across representations, so the panels are not comparable by raw value.

For intuition about what the statistic detects, Figure 3 projects the genome representations to two dimensions with PCA and colors them by phylum. Genomes fall into visibly lineage-specific groups rather than one Gaussian cloud. Two caveats keep this figure honest. First, it keeps only two of hundreds of dimensions, so the apparent degree of separation is illustrative and is not a readout of the statistic, which we compute in the full space and do not use to rank the representations.

Second, we deliberately use linear PCA rather than UMAP or t -SNE: those methods fabricate striking clusters even from Gaussian or uniform noise, so they cannot serve as evidence of non-Gaussianity, whereas PCA cannot invent structure that is not present.

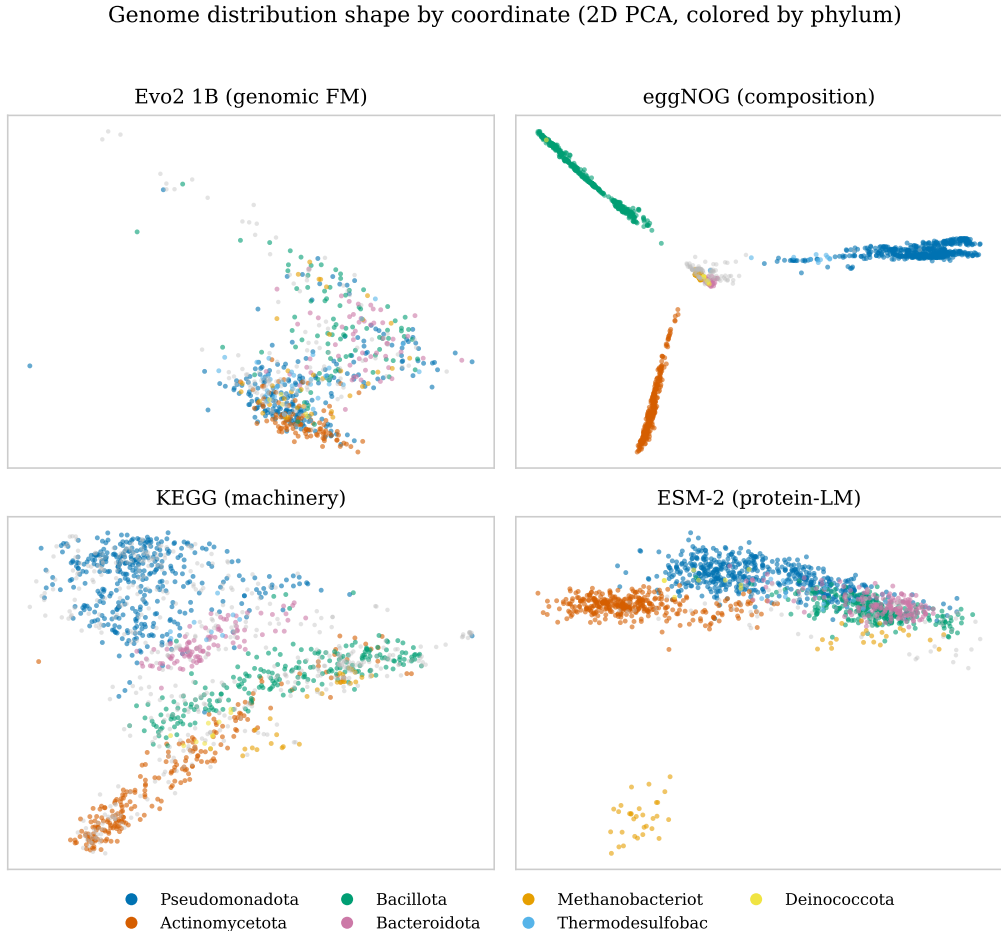


Figure 3: **What the clustering looks like (illustration only).** 2D PCA of genome representations, colored by phylum: genomes visibly organize by lineage rather than forming a single Gaussian blob. This is a lossy two-dimensional projection used for intuition; the statistic (Table 2) operates in the full space, and we do not rank representations by this view. We avoid UMAP/ t -SNE because they fabricate clusters even from Gaussian data.

Is there a data regime where genomes become Gaussian, giving a real-data sign-flip? We test the natural candidate (narrowing taxonomic breadth) at matched $N=800$ and $d=40$ to remove sample-size and dimension confounds (Table 3, Figure 4). Narrowing never reaches the Gaussian level: *every* breadth, down to a single family, clears the non-Gaussian threshold decisively (z from 12 to 47). The values are non-monotonic across levels, so we do not read a trend into them; the robust observation is only that stripping away phyla merely exposes genus-level clusters beneath, and the population stays non-Gaussian at every zoom level. Two consequences follow. First, taxonomic breadth is not a usable Gaussian axis, so it cannot supply the real-data sign-flip we would need to complete the synthetic picture. Second, the non-Gaussian reading is not a fluke of one breadth: it holds at every taxonomic scale we can measure.

Table 3: Non-Gaussianity vs. taxonomic breadth (ESM-2, matched $N=800$, $d=40$). Every level is decisively non-Gaussian.

breadth	non-Gaussianity	z
all phyla	0.0044	46.5
one phylum (Pseudomonadota)	0.0023	21.8
one class (Actinomycetes)	0.0030	28.9
one order (Caryophanales)	0.0018	11.5
one family (Streptomyetaceae)	0.0030	30.2

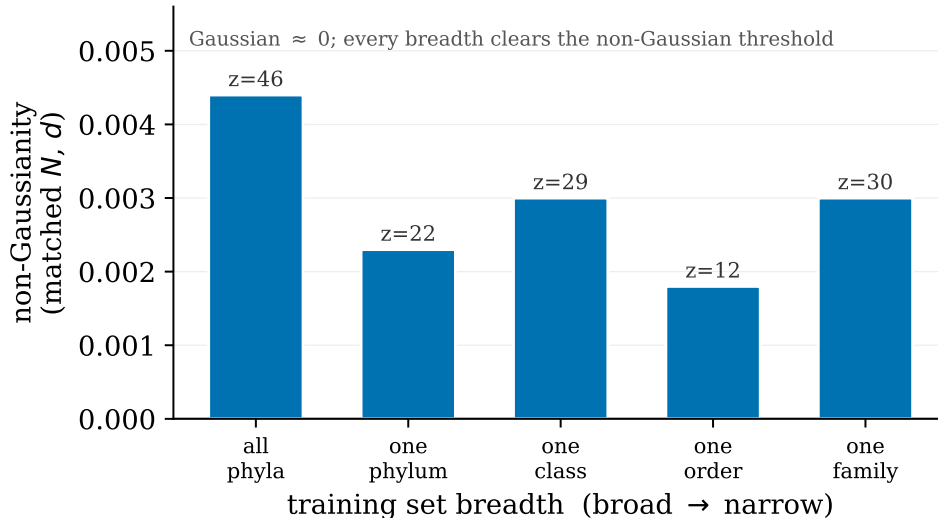


Figure 4: **Scale-invariant non-Gaussianity.** At matched N and d , every taxonomic breadth from all phyla down to a single family clears the non-Gaussian threshold ($z \gg 3$). The values are non-monotonic, so we read no trend from them; the point is only that genome clustering persists at every scale and never reaches the Gaussian level.

4.3 A second real domain is also non-Gaussian

Since genomes cluster at every scale, we asked whether a very different real domain might sit closer to the Gaussian regime, which would let us probe the other end of the selection rule (where isotropic-Gaussian is predicted to *win*). The natural candidate is vision, the setting in which isotropic-Gaussian regularization is claimed to work. We embed CIFAR-10 test images with a pretrained ResNet-18 (the vision analog of the ESM-2 coordinate) and apply the same permutation-null selector at matched $N=675$, $d=40$. The images read *decisively non-Gaussian*: real 0.0098 versus a null of 0.0010 ($z=103$; at the full $N=10,000$, $z=829$). This is unsurprising in hindsight, since CIFAR’s ten classes form clusters in a discriminative encoder, but it means an image-classification benchmark does not supply the near-Gaussian regime we sought. Both real domains we can readily access are non-Gaussian (Figure 5); finding a genuinely near-Gaussian *real* dataset, as opposed to the synthetic one of §4.1, is the key remaining obstacle to a real-data sign-flip.

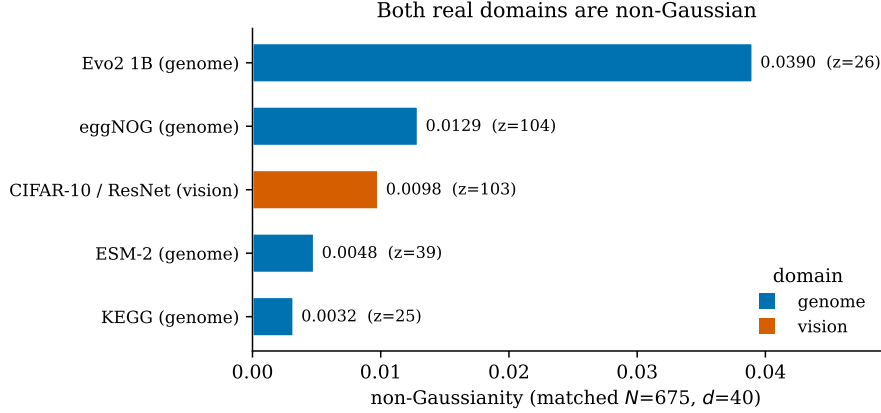


Figure 5: **Both real domains are non-Gaussian.** Non-Gaussianity of every real representation we measured, at matched $N=675$, $d=40$: four genome coordinates (blue) and vision (CIFAR-10 in a pretrained ResNet, orange). Every one clears the non-Gaussian threshold ($z \gg 3$), and vision falls squarely within the genome range. Neither accessible real domain supplies the near-Gaussian regime; only the synthetic data of §4.1 reaches it.

4.4 The crossover on real genomes

The selector places genomes in the clustered regime, so its prescription is concrete: the shape-agnostic geometries should beat the isotropic-Gaussian one on the actual genome-JEPA. We test this with a *faithful* SIGReg, the characteristic-function normality statistic of Epps–Pulley / BHEP type over random 1-D projections, which constrains *all* moments toward $\mathcal{N}(0, I)$ via Cramér–Wold rather than the first two, and we tune the regularizer loss weight *per geometry* instead of sharing one. These are the two corrections that could most plausibly have confounded an earlier, weaker version of this experiment. Table 4 and Figure 6 report the result (40 epochs, 3 seeds, best weight per geometry). The ranking matches the prediction: EMA (0.712) and VICREG (0.703) lead and SIGREG (0.684) trails. Crucially, with the faithful implementation and fair tuning SIGREG is *no longer below* the FROZEN random-feature floor (0.684 vs. 0.683); the below-floor reading of the untuned two-moment surrogate was a loss-weight artifact. The direction the selector predicts therefore survives the two fixes that were the most likely alternative explanations.

We still temper the claim. The gap between the leaders and SIGREG is small (~ 0.02 – 0.03) and rests on $n=3$ seeds without confidence intervals, so we assert a consistent *ranking*, not a large effect; FROZEN is a random-feature floor, not a fair frozen-expert; and this is a small (0.63M-parameter) genome-JEPA on 675 genomes. What the tuned, faithful run adds over the earlier version is that the ranking is not an artifact of an over-weighted or non-faithful regularizer: on non-Gaussian genome data, isotropic-Gaussian trails the shape-agnostic geometries even when implemented faithfully and tuned to its best weight.

Table 4: Powered genome bake-off with *faithful* SIGReg and *per-geometry* loss-weight tuning (best weight shown; 40 epochs, 3 seeds; per-seed values in parentheses). SIGREG no longer trails the FROZEN floor, and the shape-agnostic geometries still lead.

geometry	best reg. weight	mean AUROC	seeds
ema	n/a	0.7115	0.701, 0.726, 0.707
vicreg	0.3	0.7028	0.719, 0.687, 0.702
frozen	n/a	0.6833	0.679, 0.692, 0.679
sigreg (iso-Gaussian, faithful)	3.0	0.6835	0.689, 0.672, 0.689

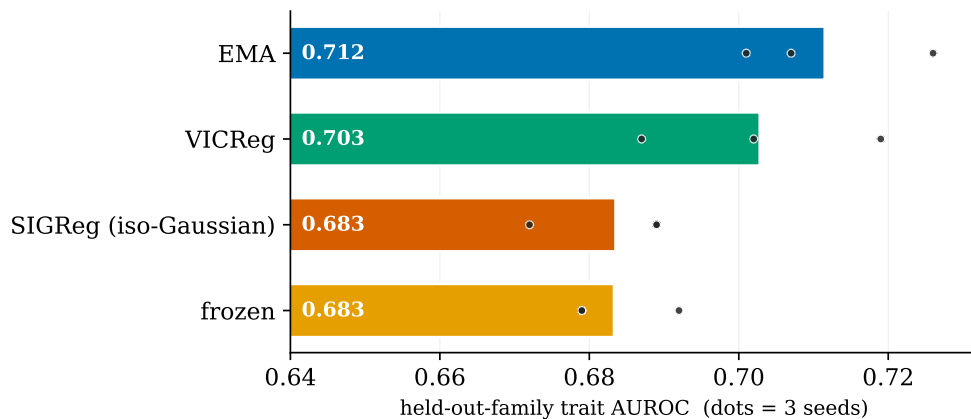


Figure 6: **The genome bake-off (faithful SIGReg, per-geometry tuning).** Held-out-family trait AUROC per geometry (bars = mean of 3 seeds; dots = seeds). The structure-preserving geometries (EMA, VICReg) lead and isotropic-Gaussian SIGReg trails, as the selector predicts; with the faithful implementation and tuned weight, SIGReg is level with the frozen floor rather than below it. The gap is small and rests on 3 seeds, so we claim a ranking, not a large effect.

5 Discussion and Limitations

What holds, and what does not. Four findings hold up. The selector, gated by a permutation null, separates genuine joint structure from per-axis and finite-sample artifacts, and its earlier random-encoder version failed exactly this test; that self-correction is part of the contribution. In controlled synthetic data the selection rule sign-flips, isotropic-Gaussian best when latents are Gaussian and worst when clustered. Genomes read non-Gaussian in every representation and at every taxonomic scale we measured. And on the genome-JEPA, with a *faithful* SIGReg and per-geometry loss-weight tuning, the shape-agnostic geometries beat isotropic-Gaussian and SIGREG no longer trails the random-feature floor (§4.4), so the ranking is not an artifact of the two things most likely to confound it. What we still do *not* establish is a large or universal effect: the genome gap is small (~ 0.02 – 0.03) and rests on three seeds, and we lack a naturally-Gaussian real dataset for the Gaussian end of the sign-flip.

The observation–latent gap. The selector is computed on data representations, not on the world’s latents, whereas the identifiability condition is on the latents (§3.2). Non-Gaussian observations are compatible with Gaussian latents, so consistency across independent representations

is suggestive but not conclusive; bridging this proxy to the latent condition is the deepest open question here.

On novelty. The synthetic sign-flip is close to what the identifiability theory already predicts and ablates [11], and the “adapt the shape rather than fix it” stance overlaps with VISReg [12]; we do not claim these as new. We do claim as new the permutation-null *validation* of a training-free selector, including the honest report that the naive version failed the CLT check, and the scale-invariance of genome non-Gaussianity.

Remaining limitations and what would settle the claim. The bake-off is $n=3$ seeds without confidence intervals, so we assert a consistent ranking rather than a large effect; selector magnitudes are small and the null level varies across representations, so we rely only on each row’s gap to its own null; FROZEN is a random-feature floor, not a fair frozen-expert; the model is small (0.63M parameters, 675 genomes); and we have one domain. Of the checks a skeptic would demand, we have now done two: the faithful characteristic-function SIGReg with per-geometry tuning (§4.4) and the selector on the JEPA’s own embeddings (§3.2). Two remain. First, a naturally-Gaussian *real* domain to supply the other end of the sign-flip; the obvious candidate, vision, is itself decisively non-Gaussian (§4.3), so identifying a genuinely near-Gaussian real dataset remains the key open obstacle. Second, ten seeds with bootstrap confidence intervals.

6 Conclusion

The regularizer a JEPA uses to avoid collapse is a design choice the field fixes a priori, and recent theory says it should depend on the data distribution. We turn that argument into a concrete, testable recipe: measure a representation’s non-Gaussianity with a permutation-validated selector, then choose the collapse-avoidance geometry accordingly. The rule sign-flips cleanly in controlled data, genomes read non-Gaussian in every representation and at every scale, and a genome bake-off ranks the geometries as the selector predicts, and that ranking survives a faithful characteristic-function SIGReg and per-geometry loss-weight tuning, the two corrections most likely to have explained it away. We are deliberate about scope: the genome effect is small and rests on three seeds, and we lack a naturally-Gaussian real control, so this is evidence that selecting geometry from the data is a promising and testable principle and that the isotropic-Gaussian default is not obviously right for genomes, not a proof that it is wrong. The remaining experiments that would settle it, chiefly a near-Gaussian real domain and more seeds, are concrete and identified above.

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